Laura Bassi Scholarship 2026/27 Application Form

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Institution	Department	Degree or Position
Ajou University	Artificial Intelligence Convergence	PhD Candidate

In 250 words or less, describe your manuscript's intended purpose. You may use this space to offer a general description of your project or an outline of your argument as it may appear in your abstract or introduction.

The manuscript reports how a distributed, dual-polarized FMCW radar detects a person behind a wall who walks across the radar's line of sight. I plan to submit it to [CONFIRM: target journal] by [CONFIRM: month, 2027].

One radar measures motion only along its own line of sight. A person who walks parallel to the wall barely changes range and produces almost no Doppler shift, so clutter suppression removes the person's echo along with the static returns. In IEEE Internet of Things Journal (2025), I placed two orthogonally polarized radar nodes apart, gave them high-gain lens array antennas, and raised the signal-to-noise ratio by polarization synthesis. At APMC 2025, I used machine learning to select the range bins for coherent integration during frontal and lateral motion. At the KIEES Winter Conference 2026, I combined the polarization channels by their band-energy difference.

The manuscript joins these steps into one processing chain and tests it on measurements through [CONFIRM: wall types, e.g. concrete] with [CONFIRM: number of subjects and motion patterns]. It compares the chain with single-radar and single-polarization processing for each motion direction, reporting detection rate and false alarms. [CONFIRM: one headline result with numbers]

Through-the-wall studies often test only a person walking toward the radar or breathing in place; this manuscript measures lateral motion. English is my second language, and editing would help me submit a clear manuscript.

In 250 words or less, describe how your field of study stands to benefit from your research.

On 11 January 2022, the upper floors of an apartment tower under construction in Hwajeong-dong, Gwangju, collapsed, and six workers went missing. A week later, five of them had still not been located; the search ended on 8 February. Search teams at such sites rely on dogs, listening devices, and cameras, which need an opening or a sound from the victim.

Radio waves pass through walls and debris, so radar can find a person who is silent and out of sight. Yet through-the-wall radar for rescue draws less research than automotive or communication radar, and most published systems use one radar with one polarization. That setup handles two cases poorly: a person who walks across the radar's line of sight, and a wall that changes the polarization of the echo.

Since 2024, in the project "Development of Core Distributed Radar Technology for Human Detection in Wide-Area Disaster Environments," I have addressed both cases with hardware I designed and measured. Two radars placed apart view the scene from different angles, so motion across one radar's line of sight moves toward the other. Dual-polarized lens array antennas receive the co- and cross-polarized echoes, which I combine by their band-energy difference. Machine-learning range-bin selection then decides where to integrate the echo coherently.

The field would gain measured results for lateral motion behind real walls, a reference that other groups can test their systems against.

SELECT THE ROUND OF FUNDING FOR WHICH YOU ARE APPLYING:

☐	SPRING 2026	DEADLINE:	8 MARCH 2026
		RESULTS:	10 APRIL 2026
☐	SUMMER 2026	DEADLINE:	12 JULY 2026
		RESULTS:	24 JULY 2026
☑	WINTER 2026	DEADLINE:	15 NOVEMBER 2026
		RESULTS:	4 DECEMBER 2026

EUN SUNG JANG — SIGNATURE OF APPLICANT

2026.10.21 — DATE

Applicants are required to submit this application form along with their curriculum vitae through the application portal by the relevant deadline. For more information about the Laura Bassi Scholarship and access to the application portal, please visit https://editing.press/bassi.

To help defray the Scholarship's administrative costs, applicants are subject to a voluntary USD 10.00 fee. **All applicants who are unable to pay the application fee are welcome to take advantage of the fee waiver option on the application portal**. If you wish to pay the application fee in a non-USD currency, please consult the Scholarship FAQ.